DAY 01 OF 10

Introduction & Getting Started

Meet Python, set up your tools and write your first programs

Today's agenda

1. What Python is and where it is used
2. Install Python and VS Code
3. Your first program
4. Variables and data types
5. Input, output and type conversion

Python in 10 Days
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What is Python?

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- A high-level, general-purpose programming language created by Guido van Rossum (first released in 1991)
- Interpreted: code runs line by line, no separate compile step
- Readable: the syntax looks close to plain English
- Free, open source and has a huge library ecosystem
- Used for data, AI, web, automation, testing and scripting

> If you can read English, you can read Python. That is why it is the first language for most beginners.

Where is Python used?

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Field	What you build	Popular libraries
Data analysis	Reports, dashboards, cleaning data	Pandas, NumPy, Matplotlib
Machine Learning	Prediction and classification models	Scikit-learn
Deep Learning	Image, speech and text models	TensorFlow, PyTorch
Generative AI	Chatbots, LLM apps, agents	LangChain, OpenAI / Claude SDKs
Web & APIs	Websites and backends	FastAPI, Flask, Django
Automation	Boring tasks done for you	os, requests, selenium

Setting up your computer

- Download Python 3 from python.org and run the installer
  - Tick "Add Python to PATH" on the first screen (very important on Windows)
- Install Visual Studio Code from code.visualstudio.com and add the Python extension
- No installation option: use Google Colab (colab.research.google.com) in your browser
- Check the install in a terminal:

```python
python --version
# Python 3.12.x  (any 3.9+ is fine)
```

> Use python3 instead of python if your system says 'command not found'.

Your first program

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- print() shows output on the screen
- Save the file as hello.py and run it: python hello.py
- Text must be inside quotes (single or double)
- You can also try code line by line in the Python shell: type python and press Enter

Python

```python
print("Hello, World!")
print("Welcome to Python")
```

OUTPUT

```
Hello, World!
Welcome to Python
```

Python syntax basics

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- Indentation (spaces at the start of a line) defines blocks. It is not optional
- Python is case sensitive: name and Name are different
- Comments start with # and are ignored by Python
- One statement per line (no semicolons needed)
- Statements end at the end of the line

Python

```python
# This is a comment
print("Python is fun")   # inline comment

if 5 > 2:
    print("Indented block")
```

OUTPUT

```
Python is fun
Indented block
```

Variables

- A variable is a name that stores a value
- Create one with = (assignment). No type declaration needed
- Rules: letters, digits, underscore; cannot start with a digit; no spaces
- Use meaningful snake_case names: student_name, total_marks
- A variable can be reassigned to a new value any time

Python

```python
name = "Asha"
age = 21
height = 5.6
is_student = True

print(name, age, height, is_student)
age = age + 1
print("Next year:", age)
```

OUTPUT

```
Asha 21 5.6 True
Next year: 22
```

Data types

- int: whole numbers (10, -3)
- float: decimal numbers (3.14)
- str: text ("hello")
- bool: True or False
- NoneType: None means 'no value'
- type() tells you the type of any value

Python

```
print(type(10))
print(type(3.14))
print(type("hi"))
print(type(True))
print(type(None))
```

OUTPUT

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
<class 'NoneType'>
```

Input and output

- input("message") asks the user to type something
- input() ALWAYS returns a string, even if the user types digits
- f-strings put variables inside text: f"Hello {name}"
- print() can take several values separated by commas

Python

```
name = input("Your name: ")
city = input("Your city: ")
print(f"Hi {name} from {city}!")
```

OUTPUT

```
Your name: Ravi
Your city: Pune
Hi Ravi from Pune!
```

Type conversion

- int(), float(), str() convert one type to another
- Needed because input() gives text
- int("abc") causes a ValueError. Only convert valid numbers
- "5" + "5" joins text, 5 + 5 adds numbers

Python

```
age = input("Age: ")        # "20" (a string)
next_age = int(age) + 1
print("Next year:", next_age)

print(int("42") + 8)         # 50
print(float("3.5") * 2)      # 7.0
print(str(100) + " marks")   # 100 marks
print("5" + "5")             # 55
```

OUTPUT

```
Age: 20
Next year: 21
50
7.0
100 marks
55
```

Practice time

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1. Print your name, city and favourite language on three lines
2. Ask the user for two numbers and print their sum
3. Ask for a name and age and print: "Ravi will be 26 next year"
4. Ask for the radius of a circle and print its area (use 3.14 * r * r)
5. Swap the values of two variables a and b

Challenge: ask for a temperature in Celsius and print it in Fahrenheit (F = C * 9/5 + 32)

Day 1 recap

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- Python is readable, free and used everywhere in data and AI
- print() for output, input() for input (always text)
- Variables store values; types: int, float, str, bool, None
- Convert types with int(), float(), str()

Homework: finish the practice set and install VS Code or open Colab

- Next: Operators and Strings